



## A simplified model having additive noises

- Until now, we have assumed that our system model can have a very general nonlinear state-space model form:

$$\begin{aligned} x_{k+1} &= f(x_k, u_k, w_k) \\ z_k &= h(x_k, u_k, v_k). \end{aligned}$$

- Since the SPKF must jointly model all random variables in the nonlinear dynamics, we needed to create  $\hat{x}_k^a = [\hat{x}_k^T, w_k^T, v_k^T]^T$  and corresponding  $\Sigma_{\tilde{x},k}^a$ .
- But, if the noises add linearly to the state and output equations, we can avoid the complexity of needing to create augmented state estimate and covariance.
- Here we consider that case, where we now assume the model form:

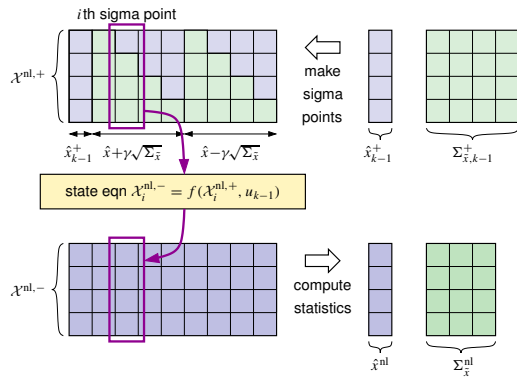
$$\begin{aligned} x_{k+1} &= f(x_k, u_k) + w_k \\ z_k &= h(x_k, u_k) + v_k. \end{aligned}$$



### Step 1a

- We seek to compute the predicted state:

$$\hat{x}_k^- = \mathbb{E}[x_k | \mathbb{Z}_{k-1}] = \underbrace{\mathbb{E}[f(x_{k-1}, u_{k-1}) | \mathbb{Z}_{k-1}]}_{\hat{x}_k^{\text{nl}}} + \underbrace{\mathbb{E}[w_{k-1} | \mathbb{Z}_{k-1}]}_{\bar{w}_{k-1}}.$$



- We use a sigma-point approach to approximate the nonlinear first term.

- Compute state-prediction sigma points:  $\mathcal{X}_i^{\text{nl},-} = f(\mathcal{X}_i^{\text{nl},+}, u_{k-1})$ .

- Compute the mean and covariance of the nonlinear first term:

$$\begin{aligned} \hat{x}_k^{\text{nl}} &= \sum_{i=0}^p \alpha_i^{(m)} \mathcal{X}_i^{\text{nl},-} \\ \Sigma_{\tilde{x},k}^{\text{nl}} &= \sum_{i=0}^p \alpha_i^{(c)} (\mathcal{X}_i^{\text{nl},-} - \hat{x}_k^{\text{nl}})(\mathcal{X}_i^{\text{nl},-} - \hat{x}_k^{\text{nl}})^T. \end{aligned}$$



### Step 1b

- Combining these results, we have:  $\hat{x}_k^- = \hat{x}_k^{\text{nl}} + \bar{w}_{k-1}$ .
- Next, to find the covariance for Step 1b.

$$\begin{aligned} \Sigma_{\tilde{x},k}^- &= \mathbb{E}[(x_k - \hat{x}_k^-)(x_k - \hat{x}_k^-)^T] \\ &= \mathbb{E}\left[(f(x_{k-1}, u_{k-1}) + w_{k-1} - \hat{x}_k^{\text{nl}} - \bar{w}_{k-1})(f(x_{k-1}, u_{k-1}) + w_{k-1} - \hat{x}_k^{\text{nl}} - \bar{w}_{k-1})^T\right] \\ &= \mathbb{E}\left[(f(x_{k-1}, u_{k-1}) - \hat{x}_k^{\text{nl}})(f(x_{k-1}, u_{k-1}) - \hat{x}_k^{\text{nl}})^T + (w_{k-1} - \bar{w}_{k-1})(w_{k-1} - \bar{w}_{k-1})^T\right. \\ &\quad \left.+ (f(x_{k-1}, u_{k-1}) - \hat{x}_k^{\text{nl}})(w_{k-1} - \bar{w}_{k-1})^T + (w_{k-1} - \bar{w}_{k-1})(f(x_{k-1}, u_{k-1}) - \hat{x}_k^{\text{nl}})^T\right]. \end{aligned}$$

- If we assume that the state is uncorrelated with present and future process noise:

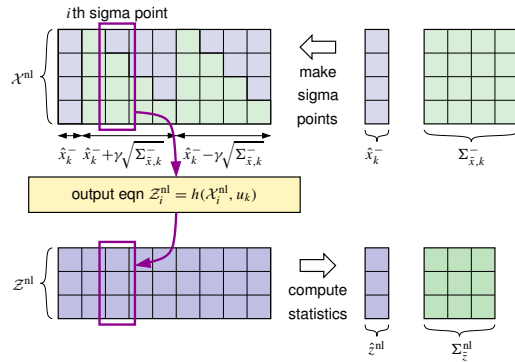
$$\begin{aligned} \Sigma_{\tilde{x},k}^- &= \mathbb{E}\left[(f(x_{k-1}, u_{k-1}) - \hat{x}_k^{\text{nl}})(f(x_{k-1}, u_{k-1}) - \hat{x}_k^{\text{nl}})^T\right] \\ &\quad + \mathbb{E}\left[(w_{k-1} - \bar{w}_{k-1})(w_{k-1} - \bar{w}_{k-1})^T\right] = \Sigma_{\tilde{x}}^{\text{nl}} + \Sigma_{\tilde{w}}. \end{aligned}$$



## Step 1c

- Now, to compute the predicted measurement:

$$\hat{z}_k = \mathbb{E}[z_k | \mathbb{Z}_{k-1}] = \underbrace{\mathbb{E}[h(x_k, u_k) | \mathbb{Z}_{k-1}]}_{\hat{z}_k^{\text{nl}}} + \underbrace{\mathbb{E}[v_k | \mathbb{Z}_{k-1}]}_{\bar{v}_{k-1}}.$$



- Existing sigma points  $\mathcal{X}_k^{\text{nl},-}$  do not contain  $\bar{w}_{k-1}$  or  $\Sigma_{\bar{w}}$  components.
- Need to generate new sigma points  $\mathcal{X}_k^{\text{nl}}$  to represent  $\hat{x}_k^-$ . Then,  $z_i^{\text{nl}} = h(\mathcal{X}_i^{\text{nl}}, u_k)$ .
- Compute the mean and covariance of the nonlinear first term:

$$\hat{z}_k^{\text{nl}} = \sum_{i=0}^p \alpha_i^{(m)} z_i^{\text{nl}}$$

$$\Sigma_{\hat{z}}^{\text{nl}} = \sum_{i=0}^p \alpha_i^{(c)} (z_i^{\text{nl}} - \hat{z}_k^{\text{nl}})(z_i^{\text{nl}} - \hat{z}_k^{\text{nl}})^T.$$



## Step 1c (continued) through 2a

- Combining these results, we have:  $\hat{z}_k = \hat{z}_k^{\text{nl}} + \bar{v}_k$ .
- For Step 2a, we need to find  $\Sigma_{\hat{z},k}$  and  $\Sigma_{\hat{x}\hat{z},k}^-$ :

$$\begin{aligned} \Sigma_{\hat{z},k} &= \mathbb{E}[(z_k - \hat{z}_k)(z_k - \hat{z}_k)^T] \\ &= \mathbb{E}\left[(h(x_k, u_k) + v_k - \hat{z}_k^{\text{nl}} - \bar{v}_k)(h(x_k, u_k) + v_k - \hat{z}_k^{\text{nl}} - \bar{v}_k)^T\right] \\ &= \mathbb{E}\left[(h(x_k, u_k) - \hat{z}_k^{\text{nl}})(h(x_k, u_k) - \hat{z}_k^{\text{nl}})^T + (v_k - \bar{v}_k)(v_k - \bar{v}_k)^T\right. \\ &\quad \left.+ (h(x_k, u_k) - \hat{z}_k^{\text{nl}})(v_k - \bar{v}_k)^T + (v_k - \bar{v}_k)(h(x_k, u_k) - \hat{z}_k^{\text{nl}})^T\right]. \end{aligned}$$

- If we assume that the state is uncorrelated with present measurement noise:

$$\begin{aligned} \Sigma_{\hat{z},k} &= \mathbb{E}\left[(h(x_k, u_k) - \hat{z}_k^{\text{nl}})(h(x_k, u_k) - \hat{z}_k^{\text{nl}})^T\right] + \mathbb{E}\left[(v_k - \bar{v}_k)(v_k - \bar{v}_k)^T\right] \\ &= \Sigma_{\hat{z}}^{\text{nl}} + \Sigma_{\bar{v}}. \end{aligned}$$



## Step 2a (continued) through 2c

- We compute  $\Sigma_{\hat{x}\hat{z},k}$  using a sigma-point approach:

$$\Sigma_{\hat{x}\hat{z},k}^- = \sum_{i=0}^p \alpha_i^{(c)} (\mathcal{X}_i^{\text{nl}} - \hat{x}_k^-)(z_i^{\text{nl}} - \hat{z}_k^{\text{nl}}).$$

- Then, the remainder of Steps 2a through 2c are standard:

$$\begin{aligned} L_k &= \Sigma_{\hat{x}\hat{z},k}^- \Sigma_{\hat{z},k}^{-1} \\ \hat{x}_k^+ &= \hat{x}_k^- + L_k(z_k - \hat{z}_k) \\ \Sigma_{\hat{x},k}^+ &= \Sigma_{\hat{x},k}^- - L_k \Sigma_{\hat{z},k} L_k^T. \end{aligned}$$



## Summary

- When noises are additive, we can implement an SPKF without augmenting the state vector before finding sigma points.
- But, now two Cholesky decompositions are needed. Is the new method faster?
  - It depends on the dimensions of  $w_k$  and  $v_k$ , speed of state and output equations.
- One choice of tuning for this method produces the “cubature Kalman filter” (CKF):

$$\gamma = \sqrt{n_x}, \quad \text{and} \quad \alpha_i^{(m)} = \alpha_i^{(c)} = \begin{cases} 0, & i = 0 \\ 1/(2n_x), & i \neq 0. \end{cases}$$

- Eliminates user-defined tuning parameters and negative  $\alpha$ ; one fewer sigma point.
- Can approximate the new method by generating  $\mathcal{X}_i^{\text{nl}} \approx \mathcal{X}_i^{\text{nl},-} + \bar{w}_{k-1}$ .
  - This avoids a second Cholesky decomposition but ignores the randomness of  $\Sigma_{\bar{w}}$  when generating  $\mathcal{X}_i^{\text{nl}}$ , which will bias the estimate  $\hat{z}_k$  and its covariance.
  - It won't be as accurate, but it might be “good enough.”